



NCI Protocol #: 10323

Patient ID:

Date:

1 of 11

Patient Name:
Date Collected:
Biopsy Site: **Liver**
Sample ID:
Telephone:
Tumor Content (%): **20_49%**

Patient DOB:
Referring Physician:
Primary Diagnosis: **Prostate Cancer**
MoCha ID:
Fax:

Tumor content assessment performed by Van Andel Research Institute; Grand Rapids; MI 49503

Table of Contents

Page

Variant Details	1
Biomarker Descriptions	2
Relevant Therapy Summary	3
Clinical Trials Summary	5

Report Highlights

2 Relevant Biomarkers
2 Therapies Available
41 Clinical Trials

Relevant Prostate Cancer Findings

Gene	Finding
AR	None detected

Relevant Biomarkers

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
III	ATM A2626V	olaparib ¹	niraparib	40
IIC	NBN splice site mutation	None	None	26

Public data sources included in relevant therapies: FDA¹, NCCN

Tier Reference: Li et al. *Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists.* J Mol Diagn. 2017 Jan;19(1):4-23.

Variant Details

DNA Sequence Variants

Gene	Amino Acid Change	Coding	Variant ID	Locus	Allele Frequency	Transcript	Variant Effect
NBN			.	chr8:90955595	68.50%	NM_002485.4	
ATM	p.(A2626V)	c.7877C>T	.	chr11:108203577	19.35%	NM_000051.3	missense

Shahanawaz Jiwani, MD, PhD, Laboratory Director | CLIA ID 21D2097127

Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2021.08(005). The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.

Comments:

N/A

Signed By: _____

Biomarker Descriptions

ATM (ATM serine/threonine kinase)

Background: The ATM gene encodes a serine/threonine kinase that belongs to the phosphatidylinositol-3-kinase related kinases (PIKKs) family of genes that also includes ATR and PRKDC (also known as DNA-PKc)¹. ATM and ATR act as master regulators of DNA damage response. Specifically, ATM is involved in double-stranded break (DSB) repair while ATR is involved in single-stranded DNA (ssDNA) repair². ATM is recruited to the DNA damage site by the MRE11/RAD50/NBN (MRN) complex that senses DSB^{2,3}. Upon activation, ATM phosphorylates several downstream proteins such as the NBN, MDC1, BRCA1, CHK2 and TP53BP1 proteins⁴. ATM is a tumor suppressor gene and loss of function mutations in ATM are implicated in the BRCAness phenotype, which is characterized by a defect in homologous recombination repair (HRR), mimicking BRCA1 or BRCA2 loss^{5,6}. Germline mutations in ATM often result in Ataxia-telangiectasia, a hereditary disease also referred to as DNA damage response syndrome that is characterized by chromosomal instability⁷.

Alterations and prevalence: Recurrent somatic mutations in ATM are observed in 17% of endometrial carcinoma, 15% of undifferentiated stomach adenocarcinoma, 13% of bladder urothelial carcinoma, 12% of colorectal adenocarcinoma, 9% of melanoma as well as esophagogastric adenocarcinoma and 8% of non-small cell lung cancer^{8,9}.

Potential relevance: The PARP inhibitor, olaparib¹⁰ is approved (2020) for metastatic castration-resistant prostate cancer (mCRPC) with deleterious or suspected deleterious, germline or somatic mutations in HRR genes that includes ATM. Consistent with other genes associated with the BRCAness phenotype, ATM mutations may aid in selecting patients likely to respond to PARP inhibitors^{5,11,12}. Specifically, in a phase II trial of metastatic, castration-resistant prostate cancer, four of six patients with germline or somatic ATM mutations demonstrated clinical responses to olaparib¹³.

NBN (nibrin)

Background: The NBN gene encodes nibrin, a nuclear protein that is part of the multisubunit MRE11/RAD50/NBN (MRN) protein complex, which is necessary for the maintenance of genomic stability^{14,15}. The MRN complex is involved in repair of double-stranded breaks (DSB) by homologous recombination repair (HRR) and non-homologous end joining (NHEJ)^{16,17,18}. Specifically, NBN contains a nuclear localization signaling motif responsible for translocation of the MRN complex into the nucleus and contributes to DNA repair by mediating protein-protein interactions at the site of DNA damage¹⁴. NBN is a tumor suppressor gene. Loss of function mutations in NBN are implicated in the BRCAness phenotype, which is characterized by a defect in the HRR pathway, mimicking BRCA1 or BRCA2 loss^{5,6}. Germline mutations in NBN are associated with Nijmegen breakage syndrome, an autosomal recessive disorder resulting in microcephaly at birth, immunodeficiency, radiosensitivity, and cancer predisposition^{19,20}.

Alterations and prevalence: Somatic mutations in NBN are observed in 7-8% of uterine cancer and 2-4% of melanoma, colorectal, esophageal, bladder and stomach cancers⁹.

Biomarker Descriptions (continued)

Potential relevance: Currently, no therapies are approved for NBN aberrations. However, loss of function mutations in one or more HRR genes, including NBN, may confer sensitivity to platinum agents and PARP inhibitors^{5,6,12}. NBN overexpression has been shown to be associated with poor prognosis in uveal melanoma, head and neck cancer, and ovarian cancer^{21,22,23,24}.

Relevant Therapy Summary

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

ATM A2626V

Relevant Therapy	FDA	NCCN	Clinical Trials*
olaparib	●	●	● (II)
niraparib	×	○	● (II)
rucaparib, hormone therapy	×	×	● (III)
talazoparib, hormone therapy	×	×	● (III)
atezolizumab	×	×	● (II)
ceralasertib	×	×	● (II)
durvalumab, olaparib	×	×	● (II)
durvalumab, tremelimumab, olaparib	×	×	● (II)
hormone therapy, olaparib, steroid	×	×	● (II)
nivolumab, chemotherapy, hormone therapy	×	×	● (II)
olaparib, chemotherapy	×	×	● (II)
olaparib, ipilimumab + nivolumab, atezolizumab + talazoparib	×	×	● (II)
olaparib, pembrolizumab	×	×	● (II)
pembrolizumab	×	×	● (II)
rucaparib	×	×	● (II)
rucaparib, atezolizumab	×	×	● (II)
rucaparib, chemotherapy	×	×	● (II)
talazoparib	×	×	● (II)
APG-115, pembrolizumab	×	×	● (I/II)
BAY-1895344	×	×	● (I/II)
copanlisib, rucaparib	×	×	● (I/II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

ATM A2626V (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
RP-3500, talazoparib	×	×	● (I/II)
talazoparib, chemotherapy	×	×	● (I/II)
BAY-1895344, pembrolizumab	×	×	● (I)
copanlisib, olaparib, durvalumab	×	×	● (I)
HH-CYH33, olaparib	×	×	● (I)
HWH-340	×	×	● (I)
olaparib, adavosertib	×	×	● (I)
talazoparib, palbociclib, axitinib, crizotinib	×	×	● (I)

NBN splice site mutation

Relevant Therapy	FDA	NCCN	Clinical Trials*
talazoparib, hormone therapy	×	×	● (III)
atezolizumab	×	×	● (II)
atezolizumab + talazoparib	×	×	● (II)
durvalumab, olaparib	×	×	● (II)
durvalumab, tremelimumab, olaparib	×	×	● (II)
niraparib	×	×	● (II)
nivolumab, chemotherapy, hormone therapy	×	×	● (II)
olaparib	×	×	● (II)
olaparib, chemotherapy	×	×	● (II)
pembrolizumab	×	×	● (II)
rucaparib	×	×	● (II)
rucaparib, atezolizumab	×	×	● (II)
talazoparib	×	×	● (II)
BAY-1895344	×	×	● (I/II)
RP-3500, talazoparib	×	×	● (I/II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Shahanawaz Jiwani, MD, PhD, Laboratory Director | CLIA ID 21D2097127

Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2021.08(005).

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

NBN splice site mutation (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
talazoparib, chemotherapy	×	×	● (I/II)
BAY-1895344, pembrolizumab	×	×	● (I)
talazoparib, palbociclib, axitinib, crizotinib	×	×	● (I)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Clinical Trials Summary

ATM A2626V

NCT ID	Title	Phase
NCT03682289	Phase II Trial of AZD6738 Alone and in Combination With Olaparib in Patients With Selected Solid Tumor Malignancies	II
NCT04564027	A Modular Phase IIa Multicentre Open-Label Study to Investigate DNA-damage Response Agents (or Combinations) in Patients With Advanced Cancer Whose Tumours Contain Molecular Alterations (PLANETTE)	II
NCT03810105	A Phase II Study of Olaparib and Durvalumab in Men With Castration Sensitive Biochemically Recurrent Non-Metastatic Prostate Cancer Harboring Mutations in DNA Damage Repair	II
NCT04030559	Phase II Trial of PARP Inhibitor Niraparib for Men With High Risk Prostate Cancer and DNA Damage Response Defects	II
NCT03533946	A Phase II Study of Rucaparib Monotherapy in Nonmetastatic, Hormone-Sensitive Prostate Cancer Demonstrating "BRCAness" Genotype (ROAR)	II
NCT04276376	A Multicenter, Open Label, Phase II Basket Trial Exploring the Efficacy And Safety of The Combination of Rucaparib (PARP Inhibitor) And Atezolizumab (Anti-PD-L1 Antibody) In Patients With DNA Repair-Deficient or Platinum-Sensitive Solid Tumors	II
NCT04586335	Open Label, Phase Ib Study to Evaluate the Safety, Tolerability, Pharmacokinetics and Clinical Activity of CYH33, an Oral a-specific PI3K Inhibitor in Combination With Olaparib, an Oral PARP Inhibitor in Patients With Advanced Solid Tumors.	I
NCT04169841	Precision Medicine Phase II Study Evaluating the Efficacy of a Double Immunotherapy by Durvalumab and Tremelimumab Combined With Olaparib in Patients With Solid Cancers and Carriers of Homologous Recombination Repair Genes Mutation in Response or Stable After Olaparib Treatment	II
NCT04126070	A Phase II Multicohort Study of Nivolumab in Combination With Docetaxel and Androgen Deprivation Therapy in Metastatic Hormone Sensitive Prostate Cancer Patients With DNA Damage Repair Defects or Inflamed Tumors	II

Clinical Trials Summary (continued)

ATM A2626V (continued)

NCT ID	Title	Phase
NCT02975934	TRITON3: A Multicenter, Randomized, Open Label Phase III Study of Rucaparib Versus Physician's Choice of Therapy for Patients With Metastatic Castration Resistant Prostate Cancer Associated With Homologous Recombination Deficiency	III
NCT04336943	Durvalumab (MEDI4736) and Olaparib (AZD2281) for Treatment of Biochemically Recurrent Prostate Cancer in Men Predicted to Have a High Neoantigen Load: A Pilot Study	II
NCT03012321	BRCAAway: A Randomized Phase II Trial of Abiraterone, Olaparib, or Abiraterone + Olaparib in Patients With Metastatic Castration-Resistant Prostate Cancer With DNA Repair Defects	II
NCT03155620	NCI-COG Pediatric MATCH (Molecular Analysis for Therapy Choice) Screening Protocol	II
NCT03742895	A Phase II Study of Olaparib Monotherapy in Participants With Previously Treated, Homologous Recombination Repair Mutation (HRRm) or Homologous Recombination Deficiency (HRD) Positive Advanced Cancer	II
NCT03967938	Efficacy of Olaparib in Advanced Cancers Occurring in Patients With Germline Mutations or Somatic Tumor Mutations in Homologous Recombination Genes	II
NCT02693535	Targeted Agent and Profiling Utilization Registry (TAPUR) Study.	II
NCT04123366	A Phase II Study of Olaparib in Combination With Pembrolizumab in Participants With Previously Treated, Homologous Recombination Repair Mutation (HRRm) and/or Homologous Recombination Deficiency (HRD)-Positive Advanced Cancer	II
NCT03248570	Phase II Open Label Study of Pembrolizumab in Patients With Metastatic Castration Resistant Prostate Cancer (mCRPC) With or Without DNA Damage Repair Defects	II
NCT03506997	PERSEUS1: Phase II Trial of the Immune Checkpoint Inhibitor Pembrolizumab for Patients Suffering From Metastatic Prostate Cancer	II
NCT02286687	Phase II Study of the PARP Inhibitor Talazoparib in Advanced Cancer Patients With Somatic Alterations in BRCA1/2, Mutations/Deletions in Other BRCA Pathway Genes and Germline Mutation in BRCA1/2 (Not Breast or Ovarian Cancer)	II
NCT02401347	A Phase II Clinical Trial of the PARP Inhibitor Talazoparib in BRCA1 and BRCA2 Wild Type Patients With Advanced Triple Negative Breast Cancer and Homologous Recombination Deficiency or Advanced HER2 Negative Breast Cancer or Other Solid Tumors With a Mutation in Homologous Recombination Pathway Genes Talazoparib Beyond BRCA (TBB) Trial	II
NCT04550494	A Pharmacodynamics-Driven Trial of Talazoparib, an Oral PARP Inhibitor, in Patients With Advanced Solid Tumors and Aberrations in Genes Involved in DNA Damage Response	II
NCT03611868	A Phase Ib/II Study of APG-115 in Combination With Pembrolizumab in Patients With Unresectable or Metastatic Melanomas or Advanced Solid Tumors	I/II
NCT03188965	An Open-label, First-in-human, Dose-escalation Study to Evaluate the Safety, Tolerability, Pharmacokinetics, Pharmacodynamics, and Maximum Tolerated Dose and / or Recommended Phase II Dose of the ATR Inhibitor BAY1895344 in Patients With Advanced Solid Tumors and Lymphomas	I/II
NCT04253262	A Phase Ib/II Study of Copanlisib Combined With Rucaparib in Patients With Metastatic Castration-resistant Prostate Cancer	I/II

Clinical Trials Summary (continued)

ATM A2626V (continued)

NCT ID	Title	Phase
NCT04095273	A Multicenter, Non-randomized, Open-label Phase Ib Study to Determine the Maximum Tolerated and Recommended Phase II Dose of the ATR Inhibitor BAY1895344 in Combination With Pembrolizumab and to Characterize Its Safety, Tolerability, Pharmacokinetics and Preliminary Anti-tumor Activity in Patients With Advanced Solid Tumors	I
NCT03842228	A Phase Ib Biomarker-Driven Combination Trial of Copanlisib, Olaparib, and Durvalumab (MEDI4736) in Patients With Advanced Solid Tumors	I
NCT04197713	Phase I Sequential Trial of Agents Against DNA Repair (STAR)	I
NCT04497116	Phase I/IIa Study of the Safety, Pharmacokinetics, Pharmacodynamics and Preliminary Clinical Activity of RP-3500 Alone or in Combination With Talazoparib in Advanced Solid Tumors With ATR Inhibitor Sensitizing Mutations (TRESR Study)	I/II
NCT03395197	A Phase III, Randomized, Double-blind, Placebo-controlled Study Of Talazoparib With Enzalutamide In Metastatic Castration-resistant Prostate Cancer	III
NCT04821622	TALAPRO-3: A PHASE III, RANDOMIZED, DOUBLE-BLIND, STUDY OF TALAZOPARIB WITH ENZALUTAMIDE VERSUS PLACEBO WITH ENZALUTAMIDE IN MEN WITH DDR GENE MUTATED METASTATIC CASTRATION-SENSITIVE PROSTATE CANCER	III
NCT03767075	Basket of Baskets: A Modular, Open-label, Phase II, Multicentre Study To Evaluate Targeted Agents in Molecularly Selected Populations With Advanced Solid Tumours	II
NCT04038502	An Open-label, Multicenter Phase II Study to Compare the Efficacy of Carboplatin as First-line Followed by Second-line Olaparib Versus Olaparib as First-line Followed by Second-line Carboplatin in the Treatment of Patients With Castration Resistant Prostate Cancer Containing Homologous Recombination Deficiency	II
NCT04901702	A Randomized Phase I/II Study of Talazoparib or Temozolomide in Combination With Onivyde in Children With Recurrent Solid Malignancies and Ewing Sarcoma	I/II
NCT04423185	Platform Study of Genotyping Guided Precision Medicine for Rare Tumors in China	II
NCT03233204	NCI-COG Pediatric MATCH (Molecular Analysis for Therapy Choice)- Phase 2 Subprotocol of Olaparib in Patients With Tumors Harboring Defects in DNA Damage Repair Genes	II
NCT03297606	Canadian Profiling and Targeted Agent Utilization Trial (CAPTUR): A Phase II Basket Trial	II
NCT03442556	PLATI-PARP: A Phase II Study of Induction Docetaxel and Carboplatin Followed by Maintenance Rucaparib in Treatment of Patients With Metastatic Castration Resistant Prostate Cancer With Homologous Recombination DNA Repair Deficiency	II
NCT03415659	A Phase I, Open-label, Single-center, Single/Multiple-dose, Dose-escalation/Dose-expansion Clinical Study on Tolerance and Pharmacokinetics of HWH340 Tablet in Patients With Advanced Solid Tumors	I
NCT04693468	Modular Phase IB Hypothesis-Testing, Biomarker-Driven, Talazoparib Combination Trial (TalaCom)	I

Clinical Trials Summary (continued)

NBN splice site mutation

NCT ID	Title	Phase
NCT04030559	Phase II Trial of PARP Inhibitor Niraparib for Men With High Risk Prostate Cancer and DNA Damage Response Defects	II
NCT03533946	A Phase II Study of Rucaparib Monotherapy in Nonmetastatic, Hormone-Sensitive Prostate Cancer Demonstrating "BRCAness" Genotype (ROAR)	II
NCT04276376	A Multicenter, Open Label, Phase II Basket Trial Exploring the Efficacy And Safety of The Combination of Rucaparib (PARP Inhibitor) And Atezolizumab (Anti-PD-L1 Antibody) In Patients With DNA Repair-Deficient or Platinum-Sensitive Solid Tumors	II
NCT03810105	A Phase II Study of Olaparib and Durvalumab in Men With Castration Sensitive Biochemically Recurrent Non-Metastatic Prostate Cancer Harboring Mutations in DNA Damage Repair	II
NCT04169841	Precision Medicine Phase II Study Evaluating the Efficacy of a Double Immunotherapy by Durvalumab and Tremelimumab Combined With Olaparib in Patients With Solid Cancers and Carriers of Homologous Recombination Repair Genes Mutation in Response or Stable After Olaparib Treatment	II
NCT04126070	A Phase II Multicohort Study of Nivolumab in Combination With Docetaxel and Androgen Deprivation Therapy in Metastatic Hormone Sensitive Prostate Cancer Patients With DNA Damage Repair Defects or Inflamed Tumors	II
NCT02693535	Targeted Agent and Profiling Utilization Registry (TAPUR) Study.	II
NCT04336943	Durvalumab (MEDI4736) and Olaparib (AZD2281) for Treatment of Biochemically Recurrent Prostate Cancer in Men Predicted to Have a High Neoantigen Load: A Pilot Study	II
NCT03012321	BRCAAway: A Randomized Phase II Trial of Abiraterone, Olaparib, or Abiraterone + Olaparib in Patients With Metastatic Castration-Resistant Prostate Cancer With DNA Repair Defects	II
NCT03967938	Efficacy of Olaparib in Advanced Cancers Occurring in Patients With Germline Mutations or Somatic Tumor Mutations in Homologous Recombination Genes	II
NCT04171700	A Phase II Multicenter, Open-label Study of Rucaparib as Treatment for Solid Tumors Associated With Deleterious Mutations in Homologous Recombination Repair Genes	II
NCT02286687	Phase II Study of the PARP Inhibitor Talazoparib in Advanced Cancer Patients With Somatic Alterations in BRCA1/2, Mutations/Deletions in Other BRCA Pathway Genes and Germline Mutation in BRCA1/2 (Not Breast or Ovarian Cancer)	II
NCT02401347	A Phase II Clinical Trial of the PARP Inhibitor Talazoparib in BRCA1 and BRCA2 Wild Type Patients With Advanced Triple Negative Breast Cancer and Homologous Recombination Deficiency or Advanced HER2 Negative Breast Cancer or Other Solid Tumors With a Mutation in Homologous Recombination Pathway Genes Talazoparib Beyond BRCA (TBB) Trial	II
NCT04550494	A Pharmacodynamics-Driven Trial of Talazoparib, an Oral PARP Inhibitor, in Patients With Advanced Solid Tumors and Aberrations in Genes Involved in DNA Damage Response	II
NCT03506997	PERSEUS1: Phase II Trial of the Immune Checkpoint Inhibitor Pembrolizumab for Patients Suffering From Metastatic Prostate Cancer	II
NCT04497116	Phase I/IIa Study of the Safety, Pharmacokinetics, Pharmacodynamics and Preliminary Clinical Activity of RP-3500 Alone or in Combination With Talazoparib in Advanced Solid Tumors With ATR Inhibitor Sensitizing Mutations (TRESR Study)	I/II

Clinical Trials Summary (continued)

NBN splice site mutation (continued)

NCT ID	Title	Phase
NCT03395197	A Phase III, Randomized, Double-blind, Placebo-controlled Study Of Talazoparib With Enzalutamide In Metastatic Castration-resistant Prostate Cancer	III
NCT04821622	TALAPRO-3: A PHASE III, RANDOMIZED, DOUBLE-BLIND, STUDY OF TALAZOPARIB WITH ENZALUTAMIDE VERSUS PLACEBO WITH ENZALUTAMIDE IN MEN WITH DDR GENE MUTATED METASTATIC CASTRATION-SENSITIVE PROSTATE CANCER	III
NCT03767075	Basket of Baskets: A Modular, Open-label, Phase II, Multicentre Study To Evaluate Targeted Agents in Molecularly Selected Populations With Advanced Solid Tumours	II
NCT04038502	An Open-label, Multicenter Phase II Study to Compare the Efficacy of Carboplatin as First-line Followed by Second-line Olaparib Versus Olaparib as First-line Followed by Second-line Carboplatin in the Treatment of Patients With Castration Resistant Prostate Cancer Containing Homologous Recombination Deficiency	II
NCT03248570	Phase II Open Label Study of Pembrolizumab in Patients With Metastatic Castration Resistant Prostate Cancer (mCRPC) With or Without DNA Damage Repair Defects	II
NCT04901702	A Randomized Phase I/II Study of Talazoparib or Temozolomide in Combination With Onivyde in Children With Recurrent Solid Malignancies and Ewing Sarcoma	I/II
NCT04095273	A Multicenter, Non-randomized, Open-label Phase Ib Study to Determine the Maximum Tolerated and Recommended Phase II Dose of the ATR Inhibitor BAY1895344 in Combination With Pembrolizumab and to Characterize Its Safety, Tolerability, Pharmacokinetics and Preliminary Anti-tumor Activity in Patients With Advanced Solid Tumors	I
NCT03233204	NCI-COG Pediatric MATCH (Molecular Analysis for Therapy Choice)- Phase 2 Subprotocol of Olaparib in Patients With Tumors Harboring Defects in DNA Damage Repair Genes	II
NCT03188965	An Open-label, First-in-human, Dose-escalation Study to Evaluate the Safety, Tolerability, Pharmacokinetics, Pharmacodynamics, and Maximum Tolerated Dose and / or Recommended Phase II Dose of the ATR Inhibitor BAY1895344 in Patients With Advanced Solid Tumors and Lymphomas	I/II
NCT04693468	Modular Phase IB Hypothesis-Testing, Biomarker-Driven, Talazoparib Combination Trial (TalaCom)	I

Assay Information

Methodology, Scope, and Application: The OCAv3 next-generation sequencing (NGS) assay identifies more than 3000 annotated mutations of interest (MOIs) broadly categorized into 5 mutation types: single nucleotide variants (SNVs), small insertions/deletions (Indels), large (>3 bases) insertions/deletions (Large Indels), copy number variants (CNVs), and gene fusions. The assay utilizes the Thermo Fisher Scientific Oncomine® Comprehensive Assay v3.0 (OCAv3), a next-generation sequencing (NGS) assay that utilizes a multiplex polymerase chain reaction (PCR) with DNA and RNA extracted from formalin-fixed tissue for sequencing on the Ion S5 XL platform and analyzed by the current version of Torrent Suite Software and Ion Reporter. The OCAv3 NGS Assay currently can reliably identify the presence or absence of >3000 known mutations and polymorphisms in the 161 unique genes compared to the Human Reference Genome hg19, including the genes listed below. The OCAv3 assay is a laboratory developed test designed to find gene mutations within tumors (somatic mutations).

Analytical Sensitivity and Specificity: The OCAv3 NGS Assay has been determined to be suitably analytically sensitive and specific for the various types of abnormalities within its reportable range, demonstrating 97.18% sensitivity compared with orthogonal assay results and 100% specificity for the 5 mutation types reported. 99.99% or greater reproducibility has been demonstrated. All quality measures for this assay were within defined assay parameters.

Hereditary and Germline Mutations: This assay examines tumor tissue only and does not examine normal (non-tumor) tissue. Mutations detected by the assay may be present only in the tumor, or in every cell of the body (including non-tumor cells). This test also cannot tell whether a potential germline mutation causes or will cause a hereditary cancer syndrome. If the patient's history includes any one or combination of features suggestive of a possible hereditary cancer predisposition (such as, cancer arising at a young age, especially a cancer of a type unusual in younger patients; a personal history of multiple different types of cancers; or a significant cancer history among blood relatives, especially but not exclusively of the same type as the patient's) it is recommended that the physician arrange for the patient to meet with a genetic counselor and, if warranted, undergo the appropriate genetic test on normal (i.e. non-tumor) tissue (blood or cells brushed from the oral surface of the cheek) to check for a germline abnormality, regardless of the results of this research study.

Report Content: Thermo Fisher Scientific's Ion Torrent OncoPrint Reporter software was used in the generation of this report. Software was developed and designed internally by Thermo Fisher Scientific. The analysis was based on the current version of OncoPrint Reporter. The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. FDA information was sourced from www.fda.gov and is current as of this version. NCCN information was sourced from www.nccn.org and reflects its current version. Clinical Trials information reflects the current version. For the most up-to-date information regarding a particular trial, search www.clinicaltrials.gov by NCT ID or search local clinical trials authority website by local identifier listed in 'Other Identifiers.' Variants are reported according to HGVS nomenclature and classified following AMP/ASCO/CAP guidelines (Li et al. 2017). Based on the data sources selected, variants, therapies, and trials identified in this report are listed in order of potential clinical significance but not for predicted efficacy of the therapies.

Genes assayed for hotspot mutations:

AKT1, AKT2, AKT3, ALK, AR, ARAF, AXL, BRAF, BTK, CBL, CCND1, CDK4, CDK6, CHEK2, CSF1R, CTNNB1, DDR2, EGFR, ERBB2, ERBB3, ERBB4, ERCC2, ESR1, EZH2, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, FOXL2, GATA2, GNA11, GNAQ, GNAS, H3F3A, HIST1H3B, HNF1A, HRAS, IDH1, IDH2, JAK1, JAK2, JAK3, KDR, KIT, KNSTRN, KRAS, MAGOH, MAP2K1, MAP2K2, MAP2K4, MAPK1, MAX, MDM4, MED12, MET, MTOR, MYC, MYCN, MYD88, NFE2L2, NRAS, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPP2R1A, PTPN11, RAC1, RAF1, RET, RHEB, RHOA, ROS1, SF3B1, SMAD4, SMO, SPOP, SRC, STAT3, TERT, TOP1, U2AF1, XPO1

Genes assayed with full exon coverage:

ARID1A, ATM, ATR, ATRX, BAP1, BRCA1, BRCA2, CDK12, CDKN1B, CDKN2A, CDKN2B, CHEK1, CREBBP, FANCA, FANCD2, FANCI, FBXW7, MLH1, MRE11, MSH2, MSH6, NBN, NF1, NF2, NOTCH1, NOTCH2, NOTCH3, PALB2, PIK3R1, PMS2, POLE, PTCH1, PTEN, RAD50, RAD51, RAD51B, RAD51C, RAD51D, RB1, RNF43, SETD2, SLX4, SMARCA4, SMARCB1, STK11, TP53, TSC1, TSC2

Genes assayed for copy number variations:

AKT1, AKT2, AKT3, ALK, AR, AXL, BRAF, CCND1, CCND2, CCND3, CCNE1, CDK2, CDK4, CDK6, EGFR, ERBB2, ESR1, FGF19, FGF3, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, IGF1R, KIT, KRAS, MDM2, MDM4, MET, MYC, MYCL, MYCN, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPARG, RICTOR, TERT

Genes assayed for detection of fusions:

AKT2, ALK, AR, AXL, BRAF, BRCA1, BRCA2, CDKN2A, EGFR, ERBB2, ERBB4, ERG, ESR1, ETV1, ETV4, ETV5, FGFR1, FGFR2, FGFR3, FGF, FLT3, JAK2, KRAS, MDM4, MET, MYB, MYBL1, NF1, NOTCH1, NOTCH4, NRG1, NTRK1, NTRK2, NTRK3, NUTM1, PDGFRA, PDGFRB, PIK3CA, PPARG, PRKACA, PRKACB, PTEN, RAD51B, RAF1, RB1, RELA, RET, ROS1, RSPO2, RSPO3, TERT

DISCLAIMER

This assay is considered a Laboratory Developed Test (LDT). Its performance characteristics have been determined through extensive testing although it has not been cleared by the US Food and Drug Administration and such approval is not required for clinical implementation. Furthermore, any comments included in the report are strictly interpretive and the opinion of the reviewer. This laboratory is certified under the Clinical Laboratory Improvements Amendments (CLIA) for the performance of high-complexity testing for clinical purposes. The correlative data presented is a result of the curation of published data sources, but may not be exhaustive. The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.

References

1. Maréchal et al. DNA damage sensing by the ATM and ATR kinases. *Cold Spring Harb Perspect Biol.* 2013 Sep 1;5(9). PMID: 24003211
2. Matsuoka et al. ATM and ATR substrate analysis reveals extensive protein networks responsive to DNA damage. *Science.* 2007 May 25;316(5828):1160-6. PMID: 17525332
3. Ditch et al. The ATM protein kinase and cellular redox signaling: beyond the DNA damage response. *Trends Biochem. Sci.* 2012 Jan;37(1):15-22. PMID: 22079189
4. Kozlov et al. Autophosphorylation and ATM activation: additional sites add to the complexity. *J. Biol. Chem.* 2011 Mar 18;286(11):9107-19. PMID: 21149446
5. Lim et al. Evaluation of the methods to identify patients who may benefit from PARP inhibitor use. *Endocr. Relat. Cancer.* 2016 Jun;23(6):R267-85. PMID: 27226207
6. Lord et al. BRCAness revisited. *Nat. Rev. Cancer.* 2016 Feb;16(2):110-20. PMID: 26775620
7. Cynthia et al. Ataxia telangiectasia: a review. *Orphanet J Rare Dis.* 2016 Nov 25;11(1):159. PMID: 27884168
8. Cerami et al. The cBio cancer genomics portal: an open platform for exploring multidimensional cancer genomics data. *Cancer Discov.* 2012 May;2(5):401-4. PMID: 22588877
9. Weinstein et al. The Cancer Genome Atlas Pan-Cancer analysis project. *Nat. Genet.* 2013 Oct;45(10):1113-20. PMID: 24071849
10. https://www.accessdata.fda.gov/drugsatfda_docs/label/2021/208558s019s020lbl.pdf
11. Gilardini et al. ATM-depletion in breast cancer cells confers sensitivity to PARP inhibition. *CR.* PMID: 24252502
12. Pennington et al. Germline and somatic mutations in homologous recombination genes predict platinum response and survival in ovarian, fallopian tube, and peritoneal carcinomas. *Clin. Cancer Res.* 2014 Feb 1;20(3):764-75. PMID: 24240112
13. Mateo et al. DNA-Repair Defects and Olaparib in Metastatic Prostate Cancer. *N. Engl. J. Med.* 2015 Oct 29;373(18):1697-708. PMID: 26510020
14. Lamarche et al. The MRN complex in double-strand break repair and telomere maintenance. *FEBS Lett.* 2010 Sep 10;584(17):3682-95. PMID: 20655309
15. Stracker et al. The MRE11 complex: starting from the ends. *Nat. Rev. Mol. Cell Biol.* 2011 Feb;12(2):90-103. PMID: 21252998
16. Bartkova et al. Aberrations of the MRE11-RAD50-NBS1 DNA damage sensor complex in human breast cancer: MRE11 as a candidate familial cancer-predisposing gene. *Mol Oncol.* 2008 Dec;2(4):296-316. PMID: 19383352
17. Rupnik et al. The MRN complex. *Curr. Biol.* 2008 Jun 3;18(11):R455-7. PMID: 18522810
18. Assenmacher et al. MRE11/RAD50/NBS1: complex activities. *Chromosoma.* 2004 Oct;113(4):157-66. PMID: 15309560
19. Chrzanowska et al. Nijmegen breakage syndrome (NBS). *Orphanet J Rare Dis.* 2012 Feb 28;7:13. doi: 10.1186/1750-1172-7-13. PMID: 22373003
20. Watanabe et al. Mutational inactivation of the nijmegen breakage syndrome gene (NBS1) in glioblastomas is associated with multiple TP53 mutations. *J. Neuropathol. Exp. Neurol.* 2009 Feb;68(2):210-5. PMID: 19151620
21. Ehlers et al. NBS1 expression as a prognostic marker in uveal melanoma. *Clin. Cancer Res.* 2005 Mar 1;11(5):1849-53. PMID: 15756009
22. Hsu et al. Identification of increased NBS1 expression as a prognostic marker of squamous cell carcinoma of the oral cavity. *Cancer Sci.* 2010 Apr;101(4):1029-37. PMID: 20175780
23. Yang et al. Increased NBS1 expression is a marker of aggressive head and neck cancer and overexpression of NBS1 contributes to transformation. *Clin. Cancer Res.* 2006 Jan 15;12(2):507-15. PMID: 16428493
24. Lee et al. Clinicopathological values of NBS1 and DNA damage response genes in epithelial ovarian cancers. *Exp. Mol. Med.* 2015 Nov 20;47:e195. PMID: 26584681